

Level 1 Training 16: Contest #2

A. Taxi Cab

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

One day, a group of N aliens arrived from Mars, and they needed a cab to get to their homes on Earth. They decided to take one cab only and divide the fair between them. The problem is that one alien thinks the best strategy is to divide the total fair evenly among them all. Another one came with an interesting way to divide the fair, and he proposed the following strategy:

- The ride will be divided into sessions, each session ends when an alien leaves the cab.
- For each session, the amount of fair in this session is divided evenly among the aliens that are still in the cab.

You're given the information of the ride, can you help each alien know what will it pay according to both strategies?

Input

The first line of input consists of an integer N ($1 \leq N \leq 1000$) – the number of aliens in the group. The next line contains N integers, the i^{th} integer ($1 \leq a_i \leq 10^8$) represents the fair when the i^{th} alien arrive to his destination – Note that the i^{th} alien arrive before the $(i + 1)^{th}$ alien for all $(i < N)$.

Output

Print two lines, each containing N numbers, the i^{th} number in the first line represents what the i^{th} alien should pay according to the *SECOND* strategy, and the i^{th} number in the second line represents what the i^{th} alien should pay according to the *FIRST* strategy. All numbers should have error value less than $1e^{-4}$.

Sample test(s)

input
3 12 25 30
output
4 10.5 15.5 10 10 10

Note

Aliens are very thin that a thousand alien can fit in one cab.

B. Chatting With Friends

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Our Friend Abdelrazeq loves online chatting. Abdelrazeq currently has n online friends on the popular social network Facebook, now he is wondering in how many ways can he create a group chat containing exactly m participants including himself?

Input

The first line of input consists of one integer T ($1 \leq T \leq 100$) — the number of test cases. Each of the next T lines consists of 2 integers n and m ($1 \leq n, m \leq 30$) — the number of Abdelrazeq's online friends and the number of participants in the group chat he wants to make.

Output

Print T lines, the i^{th} line contains one number, the number of ways to make the group chat in the i^{th} test case.

Sample test(s)

input
1 5 2
output
5

C. Blurry Vision

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Ahmed is a fresh graduated computer scientist and he started a series of applying for jobs in a great scale. Fortunately, Adobe Co. is interested in Ahmed's resume and wants to interview him at the headquarter of Adobe Co.. One of the interview questions was to implement "Average Blur" Filter. "Average Blur" Filter on an image is to make each pixel's colour value in the image equals to the average colour value of the pixel's colour value and all its adjacent pixels' colour values.

Example for "Average Blur" Filter :



let's take an example : consider this map as the original photo and each value in it points to the color value in each pixel.

15	20	33	10	13	55	2
61	72	11	49	22	43	60
99	88	41	17	50	30	89
42	5	67	35	15	73	7
36	72	8	22	75	14	26
56	11	31	10	46	83	17
42	73	61	77	3	15	19

when we apply "Average Blur" Filter on it, pixel (2,2) 's colour value will equal to the average of the pixels highlighted in red colours values. and pixel (4,7) 's colour value will equal to the average of the pixels highlighted in blue colours values. similarly, pixel (5,3) 's colour value will equal to the average of the pixels highlighted in green colours values.

15	20	33	10	13	55	2
61	72	11	49	22	43	60
99	88	41	17	50	30	89
42	5	67	35	15	73	7
36	72	8	22	75	14	26
56	11	31	10	46	83	17
42	73	61	77	3	15	19

Input

First line of the input contains a single integer represents N . ($1 \leq N \leq 1000$). a $N \times N$ map follows representing the original image pixels' colour values.

Output

Print a $N \times N$ map represents the resultant image after applying "Average Blur" Filter on the original one.

Sample test(s)

input

3
1 2 3
2 4 5
6 7 8

output

2 2 3
3 4 4
4 5 6

input

4
7 8 9 5
10 12 2 3
2 4 13 10
22 17 5 7

output

9 8 6 4
7 7 7 7
11 9 8 6
11 10 9 8

Note

All colour values must be represented in integers.

D. The Merger

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Bora3y co. has just released their latest software solution, a program they call "The Merger". The merger is a program that takes 2 images and combines them into one image. The process goes as follows:

- The program creates a new canvas with new dimensions, of course the new dimensions should be able to contain both images.
- The program draws the first image centered in the new canvas.
- The program draws the second image centered in the new canvas.

Of course some pixels may overlap, but the program manages to keep both images visible after the merging process using some secret magical algorithm. Now we need to estimate the dimensions of the obtained image after the merging process, can you help us with that?

Input

The input consists of four integers N_x , M_x , N_y and M_y ($1 \leq N_x, M_x, N_y, M_y \leq 10000$) representing the dimensions of image x and y respectively.

Output

Print two integers separated by a single space, the dimensions of the obtained image after the merging process.

Sample test(s)

input
5 10 5 10
output
5 10

input
15 55 5 100
output
15 100

Note

Both input images and the obtained image are rectangular shaped with length N_i and width M_i .

E. Divisibility

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

KOKO is a student in FCIS. A Mathematics doctor asks him to do a task.

The task is to calculate the minimum number that consists of k digits without leading zeros and is divisible by 10 and 12.

Input

The single line of the input contains integer k ($1 \leq k \leq 10^4$) which denotes number of digits.

Output

Print "-1" if there is no such number without quotes, otherwise print the minimum number.

Sample test(s)

input
3
output
120

F. Naruto Ramen

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Naruto is very fond of shadow clone Jutsu. In shadow clone Jutsu, Naruto creates multiple clones of himself. Suppose he creates N clones of himself, it means that in total there are N Narutos.

Hinata showing love towards Naruto makes his favourite dish *Ramen Ichiraku*.

Given N shadow clones of Naruto, Hinata made M dishes such that $N \leq M$.

Now Naruto wants each of his shadow clones to eat exactly one dish.



Now given the coordinates of positions where each Naruto is standing and the coordinates of dishes that Hinata made. Considering it takes 1 second for each Naruto to travel 1 unit in the coordinate axis.

He needs to find the minimum time it will take for all the Narutos to reach any of the dishes.

Since Naruto is weak in maths and lazy, he seeks your help to find the most optimal solution.

Input

First line contains number of test cases T ($1 \leq T \leq 20$), each test case contains two integers N and M ($1 \leq N, M \leq 1500$).

Next line contains N space separated integers P_i representing the coordinates of the i_{th} Naruto ($-10^9 \leq P_i \leq 10^9$).

Next line contains M space separated integers $dish_i$ representing the coordinates of the i_{th} dish ($-10^9 \leq dish_i \leq 10^9$).

Both the N numbers and the M numbers are sorted in non-decreasing order.

Output

For each test case, print the minimum time needed for all the Narutos to reach to their dishes.

Sample test(s)

input
1 3 4 -4 0 2 -1 1 2 3
output

3

input

2
4 4
2 3 3 41
-1000 5 10 100
2 2
1 10
1 1000000

output

1002
999990

G. A Piece of Cake

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Reta loves cake. Her mother wants to give her a piece of cake.

There are N pieces of cake numbered from 1 to N , Reta's mother chooses a piece, if the number of this piece is divisible either by 3 or by 5, she gives it to Reta.

Given a number N , how many pieces Reta might take?

Input

The input contains a single integer $N(1 \leq N \leq 10^{18})$.

Output

Print the number of pieces Reta might take.

Sample test(s)

input
10
output
5

input
20
output
9

input
9
output
4

H. Barcode

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

One of the types of barcodes is the ISBN-10, which contains a 10-digit sequence.

To check whether the barcode is valid or not, each digit is multiplied by its (integer) weight, descending from 10 to 1. That means that the leftmost digit is multiplied by 10, and the rightmost digit is multiplied by 1. Then all the numbers are added together, if the result is divisible by 11 the barcode is valid. The barcode is invalid otherwise.

The last digit of the barcode must range from 0 to 10 (the symbol X has value equals to 10).

Given the barcode, is it a valid one?

Input

The input contains a single 10-digit barcode, contains only digits between 0 and 9, except that the last digit can be X .

Output

Print "Valid" if the barcode is valid, otherwise print "Invalid".

Sample test(s)

input
0306406152
output
Valid

input
030640615X
output
Invalid

input
030640635X
output
Invalid

I. Area

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

You are given a right angled-triangle with three side lengths a , b and c .

Print the area of this triangle.

Input

The input contains three integers a , b and c ($1 \leq a, b, c \leq 100$).

Output

Print the area of the triangle.

Sample test(s)

input
3 4 5
output
6

input
8 10 6
output
24

J. Time Travel

time limit per test: 2 seconds

memory limit per test: 64 megabytes

input: standard input

output: standard output

Alice has discovered a time machine that has not yet tested on humans. She wants to test the machine on herself. The machine takes only two inputs, the current time of the day and the time to travel ahead, in the format of hours and minutes. Help Alice know in what time exactly will the machine transmit her.

Input

Input consists of two lines; The current time of the day, followed by the time to travel ahead, all in the format of $h: m$ ($0 \leq h \leq 23$)($0 \leq m \leq 59$).

Output

Print the time after transmission, in the format of $h: m$.

Sample test(s)

input
11:4 4:17
output
15:21